Akoizo Original Practice - Form 2

Division B | Codebusters | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) What is the plaintext?

Unit: Atbash and Caesar shifts.
Worked-case data: Cipher: Caesar with forward encryption shift 3. Ciphertext: FDW. Alphabet: A=0 through Z=25.
Separate investigation observations:
Cipher: Caesar with forward encryption shift 3.
Ciphertext: FDW.
Alphabet: A=0 through Z=25.

A. CAT

B. DOG

C. XZD

D. IGZ

Answer: _____

2. (2 points) How many binary positions encode one letter here?

Unit: Baconian and fractionated Morse systems.
Worked-case data: Decode AAAAA AAAAB AABAA under the modern 26-letter Baconian convention.

A. 3

B. 26

C. 5

D. 2

Answer: _____

3. (2 points) What plaintext results when CTO is the left column?

Unit: Columnar transposition and homophonic evidence.
Worked-case data: Rectangle: Six letters in two columns. Ciphertext pieces: CTO and ATO. Clue: The plaintext starts with CAT.
Separate investigation observations:
Rectangle: Six letters in two columns.
Ciphertext pieces: CTO and ATO.
Clue: The plaintext starts with CAT.

A. CATTWO

B. CTOATO

C. CATTOO

D. ATOC TO

Answer: _____

4. (2 points) What is removed in a patristocrat?

Unit: Monoalphabetic substitution and language evidence.
Worked-case data: A ciphertext word ABBC is proposed to mean MEET or TEAM. Which fits the equality pattern?

A. All vowels

B. Normal word boundaries

C. All repeated letters

D. The substitution key space

Answer: _____

5. (2 points) What should follow an arithmetic-error review?

Unit: Competition strategy and deliberate practice.
Worked-case data: Puzzle A offers 100 points with a 0.9 estimated solve probability in 3 minutes; B offers 200 with 0.3 probability in 6 minutes. Compare expected points per minute as a planning aid.

A. A new problem exercising the corrected skill

B. Memorize the old answer only

C. Skip all verification

D. Change every cipher family

Answer: _____

6. (2 points) What begins a deduplicated BALLOON keyword alphabet?

Unit: Monoalphabetic substitution and language evidence.
Worked-case data: A ciphertext word ABBC is proposed to mean MEET or TEAM. Which fits the equality pattern?

A. AB LNO

B. BONLA

C. BALON

D. BALLOON

Answer: _____

7. (2 points) Why can a numerically true sum still be invalid?

Unit: Xenocrypts and cryptarithms.
Worked-case data: Verify the candidate SEND = 9567, MORE = 1085, MONEY = 10652.

A. It contains a zero internally

B. It uses eight letters

C. Different letters may share a forbidden digit

D. It has any carry

Answer: _____

8. (4 points) Encode ZOO with key +3.

Answer: _____
_____
_____

9. (6 points) Solve this case and explain your reasoning: Nihilist ciphertext: 65. Key coordinate: 23. Square entry: Row 4, column 2 contains R.

Answer: _____
_____
_____

10. (4 points) Compute A's expected points per minute.

Unit: Competition strategy and deliberate practice.
Worked-case data: Puzzle A offers 100 points with a 0.9 estimated solve probability in 3 minutes; B offers 200 with 0.3 probability in 6 minutes. Compare expected points per minute as a planning aid.

Answer: _____
_____
_____

11. (4 points) Why does the most frequent symbol not prove a mapping to E?

Unit: Monoalphabetic substitution and language evidence.
Worked-case data: A ciphertext word ABBC is proposed to mean MEET or TEAM. Which fits the equality pattern?

Answer: _____
_____
_____

12. (4 points) Why is the heuristic uncertain?

Unit: Competition strategy and deliberate practice.
Worked-case data: Puzzle A offers 100 points with a 0.9 estimated solve probability in 3 minutes; B offers 200 with 0.3 probability in 6 minutes. Compare expected points per minute as a planning aid.

Answer: _____
_____
_____

13. (4 points) A decoded word is misspelled in an erristocrat. What should be checked first?

Answer: _____
_____
_____

14. (4 points) How should a guessed Spanish DE be tested?

Unit: Xenocrypts and cryptarithms.
Worked-case data: Verify the candidate SEND = 9567, MORE = 1085, MONEY = 10652.

Answer: _____
_____
_____

15. (4 points) Design a useful entry for an error log.

Unit: Cryptography foundations and the solving workflow.
Worked-case data: A = 0 and a Caesar key of +3 are used to encode CAT. Decode FDW and verify.

Answer: _____
_____
_____

16. (2 points) Define “Crib” in the context of Codebusters.

Answer: _____
_____
_____

17. (4 points) Design a short interleaved drill.

Unit: Competition strategy and deliberate practice.
Worked-case data: Puzzle A offers 100 points with a 0.9 estimated solve probability in 3 minutes; B offers 200 with 0.3 probability in 6 minutes. Compare expected points per minute as a planning aid.

Answer: _____
_____
_____

18. (2 points) Define “Plaintext” in the context of Codebusters.

Answer: _____
_____
_____